Nuclear Burning Stages of Stars: pp Chain, CNO Cycles, Shell Model and Gamma Decay

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Outline

Recap and Temperature Dependence of $p + p$

Recap: Where We Stand

From Lecture 22 we established:

- Net process $4p \to {}^4\text{He}$, $Q = 26.73$ MeV; effective $Q_{\text{eff}} = 26.2$ MeV after losing $2\langle E_\nu \rangle$ to escaping neutrinos
- First step $p + p \to d + e^+ + \nu$ proceeds via the weak interaction with $Q = 1.44$ MeV
- $\tau_{pp} \approx 10^{10}$ y at solar interior conditions – the chain's rate-limiting step
- Sun's hydrogen-burning lifetime $\approx 10^{10}$ y (10% core hydrogen $\Rightarrow$ middle-aged star)

In this set (Lectures 23–25) we now ask:

- Once d and ${}^3\text{He}$ are produced, how are they destroyed?
- What sets the relative importance of the pp-I, pp-II, and pp-III branches?
- How do CNO cycles (and their extensions) synthesize elements up to silicon?
- What tools – shell model, gamma-decay selection rules – do we need to describe the transitions involved?

Temperature Dependence of the $p + p$ Reaction

For a non-resonant charged-particle reaction:

$$\varepsilon \propto \langle \sigma v \rangle_{pp} \propto T^{-2/3}\, e^{-\tau}, \qquad \tau = \frac{3E_0}{kT} = 42.46\,(Z_1^2 Z_2^2\, \mu / T_6)^{1/3}$$

Plotting ε vs. T_6 for $p + p$ vs. the CNO cycle reveals a crossover at $T_6 \approx 20$:

- $T_6 \lesssim 20$: $p + p$ dominates energy production (Sun and lower-mass stars, $M \lesssim M_\odot$)
- $T_6 \gtrsim 20$: CNO cycle takes over (more massive main-sequence stars)

Five interlocked quantities along the main sequence:

$$T \uparrow \;\Leftrightarrow\; M \uparrow \;\Rightarrow\; \text{fuel consumption rate} \uparrow,\;\; \text{energy production} \uparrow,\;\; \text{lifetime} \downarrow$$

More massive stars consume their fuel faster $\Rightarrow$ shorter lifetimes – traceable directly through the HR diagram.

Deuterium Burning

Destruction Channels for Deuterium

Deuterium is *produced* by $p + p \to d + e^+ + \nu$. The available destruction channels (all exothermic):

Reaction	Q-value (MeV)
$p + d \to {}^3\mathrm{He} + \gamma$	5.5
$d + d \to {}^4\mathrm{He} + \gamma$	23.8
$d + d \to p + t$	4.03
$d + d \to {}^3\mathrm{He} + n$	3.27
${}^3\mathrm{He} + d \to {}^4\mathrm{He} + p$	18.35
${}^3\mathrm{He} + d \to {}^5\mathrm{Li} + \gamma$	16.38
${}^4\mathrm{He} + d \to {}^6\mathrm{Li} + p$	1.47

Because protons are by far the most abundant reactant, the dominant destruction channel is

$$p + d \to {}^3\mathrm{He} + \gamma \quad (5.5\ \mathrm{MeV}\ \gamma\ \text{signature})$$

Quasi-Equilibrium of the D/H Ratio

Time rate of change of deuterium:

$$\frac{dD}{dt} = r_{pp} - r_{pd} = \frac{H^2}{2}\langle\sigma v\rangle_{pp} - HD\langle\sigma v\rangle_{pd}$$

This is a self-regulating system reaching quasi-equilibrium ($dD/dt = 0$):

$$\boxed{\left(\frac{D}{H}\right)_e = \frac{\langle\sigma v\rangle_{pp}}{2\langle\sigma v\rangle_{pd}}}$$

Interpretation of the strength hierarchy:

- $p + p$ is a weak-interaction reaction $\Rightarrow$ slow production
- $p + d$ is an EM-interaction reaction $\Rightarrow$ fast destruction
- Numerically: $(D/H)_e \approx 10^{-18}$ at $T_6 = 5$, i.e. $\ll 1$

Lifetimes reflect the same hierarchy:

$$\tau_H(\text{against } p) \approx 10^{10}\ \text{y}, \qquad \tau_d(\text{against } p) \approx 1.6\ \text{s}$$

The Observed D/H Puzzle and ^{3}He's Minor Role

Puzzle: the equilibrium prediction gives $(D/H)_e \sim 10^{-18}$, but the observed ratio is

$$(D/H)_{\mathrm{obs}} \approx 10^{-5}$$

That is a 13-order-of-magnitude mismatch – an active question in nuclear astrophysics, pointing to deuterium production *prior to* stellar formation (Big Bang nucleosynthesis). **Why ^{3}He plays no role in destroying d:**

$$\tau_{^3\mathrm{He}}(d) \approx 10^4 \text{ s} \quad \text{vs.} \quad \tau_d(p) \approx 1.6 \text{ s}$$

Although $\langle \sigma v \rangle_{^3\mathrm{He},d}$ is *seven* orders of magnitude larger than $\langle \sigma v \rangle_{pd}$, the lifetime against ^{3}He is four orders of magnitude *longer* – because the ^{3}He density is so low. $\Rightarrow$ The destruction of d is dominated entirely by $p + d \to {}^3\mathrm{He} + \gamma$.

^{3}He Burning and Completion of pp-I

Destruction Channels for ^{3}He

^{3}He is produced by $p + d \to {}^3\text{He} + \gamma$. Possible destruction reactions:

Reaction	Q-value (MeV)
$d + {}^3\text{He} \to {}^5\text{Li} + \gamma$	16.38
$d + {}^3\text{He} \to {}^4\text{He} + p$	18.35
${}^3\text{He} + {}^3\text{He} \to {}^6\text{Be} + \gamma$	11.50
${}^3\text{He} + {}^3\text{He} \to {}^4\text{He} + 2p$	12.86
${}^4\text{He} + {}^3\text{He} \to {}^7\text{Be} + \gamma$	1.58

Selecting the dominant channel: extrapolated $S(0)$ values show $d + {}^3\text{He}$ and ${}^3\text{He} + {}^3\text{He} \to {}^4\text{He} + 2p$ are the key contenders. But $[d] \ll [{}^3\text{He}]$, so

$$\boxed{{}^3\text{He} + {}^3\text{He} \to {}^4\text{He} + 2p \;\text{ dominates destruction of } {}^3\text{He}}$$

Why $p + {}^3$He Does Not Complete the Chain

A natural question: protons are abundant – why not

$$^3\text{He}(p,\gamma)^4\text{Li} \xrightarrow{\beta^+} {}^4\text{He} \ ?$$

Answer: the Q-value of ^{3}He$(p,\gamma)^4$Li is -2.5 MeV (endothermic), and ^{4}Li is not particle-stable. The reaction cannot proceed under stellar conditions. **Therefore the pp-I chain completes via:**

$$p + p \to d + e^+ + \nu$$
$$p + d \to {}^3\text{He} + \gamma$$
$$^3\text{He} + {}^3\text{He} \to {}^4\text{He} + 2p$$

Net: $4p \to {}^4\text{He} + 2e^+ + 2\nu$ with $Q = 26.73$ MeV. This pp-I chain operates in the cores of

first-generation stars and in low-mass second-generation stars ($M \lesssim M_\odot$).

Equilibrium ^{3}He Abundance

$$\frac{d(^3\mathrm{He})}{dt} = r_{pd} - r_{33} = HD\,\langle\sigma v\rangle_{12} - \frac{(^3\mathrm{He})^2}{2}\,2\langle\sigma v\rangle_{33}$$

(Factor 2 in numerator: two ^{3}He per fusion. Factor 2 in denominator: identical-particle correction.)
Substituting $(D/H)_e = \langle\sigma v\rangle_{pp}/[2\langle\sigma v\rangle_{pd}]$:

$$\frac{d(^3\mathrm{He})}{dt} = \frac{H^2\langle\sigma v\rangle_{11}}{2} - (^3\mathrm{He})^2\langle\sigma v\rangle_{33}$$

Quasi-equilibrium ($d(^3\mathrm{He})/dt = 0$):

$$\boxed{\left(\frac{^3\mathrm{He}}{H}\right)_e = \left(\frac{\langle\sigma v\rangle_{11}}{2\langle\sigma v\rangle_{33}}\right)^{1/2}}$$

Lifetimes at $X_H = X_{He} = 0.5$, $\rho = 100$ g/cm^3:

$$\tau_{^3\mathrm{He}}(^3\mathrm{He}) \approx 10^5\ \mathrm{y}, \qquad \tau_d(p) \approx 1.6\ \mathrm{s}$$

$\Rightarrow$ deuterium plays *no role* in destroying ^{3}He – ^{3}He + ^{3}He dominates.

Energy Production via the $p + p$ Reaction

Energy production depends on whether ^{3}He has reached equilibrium. The overall rate is set by the slow $p + p$ step. **Step-by-step energy bookkeeping:**

- Fusion of *3* protons to ^{3}He: $Q = 6.936$ MeV, average $\langle E_\nu \rangle = 0.263$ MeV

$$E_{\text{eff}}(3p \rightarrow {}^3\text{He}) = 6.673 \text{ MeV}$$

- Fusion of *4* protons to ^{4}He via $^3\text{He} + {}^3\text{He}$: $Q = 12.860$ MeV

Total energy production rate: at equilibrium ($2r_{33} = r_{11}$),

$$\boxed{\varepsilon_{pp} = 13.103\, r_{11} \text{ MeV cm}^{-3}\,\text{s}^{-1}}$$

The overall reaction rate is entirely set by $r_{11} = r_{pp}$, so we may equivalently write $\varepsilon \approx 13 \times r_{pp}$.

The pp-II and pp-III Chains

Time for ^{3}He to Reach Equilibrium

Define f = fraction of ^{3}He abundance at time t compared to equilibrium. Let t_f = time required to reach 99% of equilibrium ($f = 0.99$). At $X_H = X_{He} = 0.5$, $\rho = 100$ g/cm^3:

- $T_6 = 5$: $t_f \sim 10^{25}$ y (longer than any star will ever live)
- $T_6 = 15$ (solar core): $t_f \sim 10^6$ y – comparable to early evolutionary timescales
- $T_6 = 30$: $t_f \sim 10^5$ y

Consequence: the ^{3}He abundance *depends on the age of the star*. For the Sun, only after $\sim 10^6$ years did enough ^{3}He accumulate for $^3\mathrm{He}(^3\mathrm{He}, 2p)^4\mathrm{He}$ to operate at its maximum rate. This is why energy-production calculations must distinguish between the pre-equilibrium and equilibrium regimes.

The Three Branches of the pp Chain

After $p + d \to {}^3\text{He} + \gamma$, the ^{3}He has two possible fates:

- **pp-I (86%):** ${}^3\text{He} + {}^3\text{He} \to {}^4\text{He} + 2p$
  - $Q_{\text{eff}} = 26.2$ MeV (2% loss to ν)
- **pp-II / pp-III (14%):** ${}^3\text{He} + {}^4\text{He} \to {}^7\text{Be} + \gamma$, then ^{7}Be branches:

pp-II: ${}^7\text{Be} + e^- \to {}^7\text{Li} + \nu$, ${}^7\text{Li} + p \to 2\,{}^4\text{He}$
$Q_{\text{eff}} = 25.66$ MeV (4% ν loss) **pp-III:** ${}^7\text{Be} + p \to {}^8\text{B} + \gamma$, ${}^8\text{B} \xrightarrow{\beta^+} {}^8\text{Be}^* \to 2\,{}^4\text{He}$
$Q_{\text{eff}} = 19.17$ MeV (28.3% ν loss – largest) In pp-II and pp-III, ^{4}He acts as a catalyst: an alpha enters and (effectively) two alphas come out, with 4 protons converted to ^{4}He overall.

Time-Evolution Equations for the pp Chains

The full set of coupled ODEs describing the abundances:

$$
\begin{aligned}
\frac{d(H)}{dt} &= -2\lambda_{11}\frac{H^2}{2} - \lambda_{12}HD + 2\lambda_{33}\frac{(^3\mathrm{He})^2}{2} - \lambda_{17}H(^7\mathrm{Be}) - \lambda_{17}^{*}H(^7\mathrm{Li}) \\
\frac{d(D)}{dt} &= \lambda_{11}\frac{H^2}{2} - \lambda_{12}HD \\
\frac{d(^3\mathrm{He})}{dt} &= \lambda_{12}HD - 2\lambda_{33}\frac{(^3\mathrm{He})^2}{2} - \lambda_{34}(^3\mathrm{He})(^4\mathrm{He}) \\
\frac{d(^4\mathrm{He})}{dt} &= \lambda_{33}\frac{(^3\mathrm{He})^2}{2} - \lambda_{34}(^3\mathrm{He})(^4\mathrm{He}) + 2\lambda_{17}H(^7\mathrm{Be}) + 2\lambda_{17}^{*}H(^7\mathrm{Li}) \\
\frac{d(^7\mathrm{Be})}{dt} &= \lambda_{34}(^3\mathrm{He})(^4\mathrm{He}) - \lambda_{e7}n_e(^7\mathrm{Be}) - \lambda_{17}H(^7\mathrm{Be}) \\
\frac{d(^7\mathrm{Li})}{dt} &= \lambda_{e7}n_e(^7\mathrm{Be}) - \lambda_{17}^{*}H(^7\mathrm{Li})
\end{aligned}
$$

Each λ_{ij} is a reaction rate; minus signs $\Rightarrow$ destruction, plus signs $\Rightarrow$ production.

Net Energy Production in the Sun

Solving the abundance equations under solar conditions:

- Total energy production rate (per gram): $\varepsilon_{\text{tot}} = 5.1 \times 10^7$ MeV g^{-1} s^{-1}
- Observed solar luminosity: $L_\odot = 2.4 \times 10^{39}$ MeV s^{-1}

Mass of hydrogen actively burning:

$$m_\odot^{\text{burning}} = \frac{L_\odot}{\varepsilon_{\text{tot}}} = 4.7 \times 10^{31}\ \text{g}$$

Compared to $M_\odot = 2 \times 10^{33}$ g:

$$\boxed{\text{Only } \approx 2.4\% \text{ of the Sun's mass is actively burning hydrogen via pp chain}}$$

This tiny fraction is enough to power the Sun for $\sim 10^{10}$ years.

The CN Cycle

Why a Different Cycle for More Massive Stars?

Most present-day stars are second- and third-generation (Population I) – their gas was enriched by earlier stellar generations with elements heavier than H, He. **Selecting the next reactant:** we need an abundant nucleus with the lowest Coulomb barrier.

- Li, Be, B: very low abundance (the famous gap in the abundance curve) – excluded
- C, N: comparatively abundant; among these, ^{12}C has the lowest Coulomb barrier

$\Rightarrow$ The next set of reactions begins with $^{12}\mathrm{C}(p,\gamma)^{13}\mathrm{N}$, opening the CN cycle.

The CN Cycle

$$^{12}\mathrm{C}(p,\gamma)^{13}\mathrm{N} \xrightarrow{\beta^+} {}^{13}\mathrm{C}(p,\gamma)^{14}\mathrm{N}(p,\gamma)^{15}\mathrm{O} \xrightarrow{\beta^+} {}^{15}\mathrm{N}(p,\alpha)^{12}\mathrm{C}$$

Key features:

- ^{12}C is consumed and regenerated $\Rightarrow$ it acts as a catalyst
- 4 protons are consumed; net reaction is again $4p \rightarrow {}^{4}\mathrm{He}$, $Q = 26.73$ MeV
- The cycle proceeds via (p,γ) and (p,α) reactions plus β^+ decays

Rate-limiting step: governed by the *slowest* reaction = the one with the highest Coulomb barrier proceeding via EM force.

$$^{14}\mathrm{N}(p,\gamma)^{15}\mathrm{O}\ \text{(EM force, slowest)}\ \text{vs.}\ {}^{15}\mathrm{N}(p,\alpha)^{12}\mathrm{C}\ \text{(strong force, fast)}$$

Significance: the CN cycle is not the primary energy source in the Sun (solar core T too low to overcome the higher Coulomb barriers), but it is essential for the synthesis of C, N isotopes – and it dominates energy production in more massive stars.

CNO Bi-cycle and Additional Cycles

The CNO Bi-cycle: Adding the Branch at ^{15}N

In the CN cycle we wrote $^{15}\mathrm{N}(p,\alpha)^{12}\mathrm{C}$. But ^{15}N can also undergo

$$^{15}\mathrm{N}(p,\gamma)^{16}\mathrm{O}(p,\gamma)^{17}\mathrm{F} \xrightarrow{\beta^+} {}^{17}\mathrm{O}(p,\alpha)^{14}\mathrm{N}$$

This is the second cycle – together with the first they form the CNO bi-cycle. **Relative importance of the two branches at ^{15}N:**

$$^{15}\mathrm{N} + p \to {}^{12}\mathrm{C} + \alpha \quad (Q = 4.965\ \mathrm{MeV},\ \mathrm{strong})$$
$$^{15}\mathrm{N} + p \to {}^{16}\mathrm{O} + \gamma \quad (Q = 12.126\ \mathrm{MeV},\ \mathrm{EM})$$

Ratio of $S(0)$ values: $\sim 1000:1$ in favor of (p,α). $\Rightarrow$ Only about 1 in 1000 reactions follows the second branch – the bi-cycle's second loop contributes very little to the total energy production, but is essential for synthesizing heavier C/N/O isotopes.

Additional Branches: CNO3 and CNO4

The story extends further whenever ^{17}O can react via (p, γ):

$$^{17}\mathrm{O}(p,\gamma)^{18}\mathrm{F} \xrightarrow{\beta^+} {}^{18}\mathrm{O}(p,\alpha)^{15}\mathrm{N}$$

and via the fourth branch:

$$^{18}\mathrm{O}(p,\gamma)^{19}\mathrm{F}, \qquad {}^{19}\mathrm{F}(p,\alpha)^{16}\mathrm{O} \text{ or } {}^{19}\mathrm{F}(p,\gamma)^{20}\mathrm{Ne}$$

Branching at ^{18}O: the ratio of (p, α) vs. (p, γ) depends strongly on temperature:

- At $T_9 < 0.02$ or $T_9 > 0.7$: hydrogen burning of ^{18}O proceeds entirely via (p, α)
- In between: (p, γ) contributes

^{19}F branch: if $^{19}\mathrm{F}(p,\alpha)^{16}\mathrm{O}$ dominates, CNO catalysts are retained (CNO4); if $^{19}\mathrm{F}(p,\gamma)^{20}\mathrm{Ne}$ dominates, the cycle is broken and ^{20}Ne seeds the next cycle (NeNa). $\Rightarrow$ Four interconnected branches: CNO1, CNO2, CNO3, CNO4 – each fusing 4 protons to ^{4}He but synthesizing different stable isotopes.

The "Many-Cycled Monster": Hot CNO at High T

In novae and supernovae ($T \gtrsim 10^9$ K, $M/M_\odot = 10^4$–10^8), the CNO cycle operates on a timescale of seconds. Under these conditions, β^+-unstable nuclei (^{13}N, ^{17}F, ^{18}F, ^{19}Ne) *react with protons before they can decay*. **New pathways open up at high T:**

- $^{13}\mathrm{N}(p,\gamma)^{14}\mathrm{O} \xrightarrow{\beta^+} {}^{14}\mathrm{N}$ (bypasses $^{13}\mathrm{N} \rightarrow {}^{13}\mathrm{C}$)
- $^{17}\mathrm{F}(p,\gamma)^{18}\mathrm{Ne} \xrightarrow{\beta^+} {}^{18}\mathrm{F} \rightarrow {}^{15}\mathrm{O}$
- $^{19}\mathrm{Ne}(p,\gamma)^{20}\mathrm{Na} \xrightarrow{\beta^+} {}^{20}\mathrm{Ne}$

This is the hot CNO regime – a richly interconnected network combining (p,γ), (p,α), and β^+ pathways. It is the gateway to heavier-element nucleosynthesis in explosive stellar environments.

NeNa and MgAl Cycles

The Neon–Sodium (NeNa) Cycle

Once ^{20}Ne is produced (from ^{19}F(p,γ)), it seeds a new cycle:

$$^{20}\mathrm{Ne}(p,\gamma)^{21}\mathrm{Na} \xrightarrow{\beta^+} {}^{21}\mathrm{Ne}(p,\gamma)^{22}\mathrm{Na}$$
$$\xrightarrow{\beta^+} {}^{22}\mathrm{Ne}(p,\gamma)^{23}\mathrm{Na} \xrightarrow{(p,\alpha)} {}^{20}\mathrm{Ne}$$

^{20}Ne acts as catalyst $\Rightarrow$ another $4p \to {}^{4}$He cycle. **Why this cycle matters:**

- Not important for energy production – the Coulomb barriers are too high
- Crucial for the synthesis of Ne and Na isotopes between $A = 20$ and $A = 23$
- ^{22}Ne is detected in meteorites $\Rightarrow$ direct laboratory evidence of stellar nucleosynthesis

The Magnesium–Aluminium (MgAl) Cycle

The NeNa cycle's branch $^{23}\mathrm{Na}(p,\gamma)^{24}\mathrm{Mg}$ initiates the MgAl cycle:

$$^{24}\mathrm{Mg}(p,\gamma)^{25}\mathrm{Al} \xrightarrow{\beta^+} {}^{25}\mathrm{Mg}(p,\gamma)^{26}\mathrm{Al}$$
$$\xrightarrow{\beta^+} {}^{26}\mathrm{Mg}(p,\gamma)^{27}\mathrm{Al} \xrightarrow{(p,\alpha)} {}^{24}\mathrm{Mg}$$

$^{26}\mathrm{Al}$ may also undergo (p,γ) to $^{27}\mathrm{Si} \xrightarrow{\beta^+} {}^{27}\mathrm{Al}$, which can choose between $(p,\alpha) \to {}^{24}\mathrm{Mg}$ (cycle closes) and $(p,\gamma) \to {}^{28}\mathrm{Si}$ (cycle exits to silicon burning). **Significance:**

- $^{26}\mathrm{Al}$ is found in meteorites – a famous cosmochronometer ($\tau_{1/2} \approx 7.2 \times 10^5$ y)
- Provides isotopes between Ne and Al
- Eventual exit at $^{28}\mathrm{Si}$ leads into silicon burning $\Rightarrow$ iron-peak elements
- Beyond iron, synthesis must proceed by neutron capture (*s*- and *r*-processes)

Nuclear Shell Model

Why We Need the Shell Model

The liquid-drop model successfully accounts for binding energies and fission, but *cannot* explain:

- The magic numbers (N or $Z = 2, 8, 20, 28, 50, 82, 126$) of unusually stable nuclei
- The existence and ordering of discrete nuclear excited states

Shell model advantages:

- Predicts properties of both closed and partially filled shells
- Provides the spin J and parity π of nuclear states
- Enables calculation of γ-ray transition strengths, spectroscopic factors, and weak-interaction matrix elements

Why this matters in nuclear astrophysics: every (p, γ) reaction populates a compound nucleus in an excited state. To predict its decay (gamma emission, branching ratios) we need J^π values and transition strengths – inputs that feed directly into stellar evolution codes.

Shell Model Essentials

Differences from the atomic shell model:

- No precise nuclear potential (atomic case: Coulomb is exact)
- Two species of fermions (protons and neutrons), not one (electrons)
- No heavy central force – nucleons move in their mutually generated potential

Solving the Schrödinger equation with a harmonic-oscillator or Woods–Saxon potential plus strong spin–orbit coupling gives single-particle states labelled by:

$$(n,\, l,\, j) \qquad \text{with} \qquad |l - 1/2| \leq j \leq l + 1/2$$

Filling rules:

- Each state of given j accommodates a maximum of $(2j + 1)$ identical nucleons (Pauli)
- Parity: $\pi = (-1)^l$ – $l = 0$ (s), 2 (d), 4 (g) $\Rightarrow$ even parity; $l = 1$ (p), 3 (f) $\Rightarrow$ odd parity
- Notation: $n\, l_j$ where $l = 0, 1, 2, 3, 4 \leftrightarrow$ s, p, d, f, g

Filling shells reproduces the magic numbers 2, 8, 20, 28, 50, 82, 126.

Gamma Decay

Gamma Decay: Energetics

Each nuclear state has a specific excitation energy E_x, populated by capture reactions, β decay, Coulomb excitation, or scattering. The dominant de-excitation is gamma decay; internal conversion and internal pair production are subleading (pair production requires $E_x > 1.02$ MeV). **Energy conservation:**

$$E_\gamma = E_i - E_f - E_{\text{recoil}}$$

Recoil kinetic energy is typically $\sim$ eV $\Rightarrow$ negligible, so

$$\boxed{E_\gamma \approx E_i - E_f}$$

X-rays vs. gamma rays: same electromagnetic radiation, different origin:

- X-rays: atomic electron rearrangements
- Gamma rays: nuclear transitions

Lab example: $^{137}\text{Cs} \xrightarrow{\beta^-} {}^{137}\text{Ba}^*$ (excited) $\xrightarrow{\gamma} {}^{137}\text{Ba}$ (ground) – gamma emission is essentially always preceded by β decay.

Gamma Decay: Angular Momentum and Multipolarity

The photon must conserve both angular momentum and parity. Define the multipolarity:

$$|I_f - I_i| \leq l \leq I_f + I_i \qquad \text{(vector coupling)}$$

The change in intrinsic angular momentum is $\Delta\hbar = (I_f - I_i)\hbar$. The emitted photon's multipolarity is l, with radiation pattern 2^l-pole:

$$l = 1 \;\; \text{(dipole)}, \qquad l = 2 \;\; \text{(quadrupole)}, \qquad l = 3 \;\; \text{(octupole)}, \ldots$$

Electric vs. magnetic character (governed by parity change):

$$\Delta\pi = (-1)^l \;\Rightarrow\; \text{electric multipole} \qquad \Delta\pi = (-1)^{l+1} \;\Rightarrow\; \text{magnetic multipole}$$

Type	Nature	l	$\Delta\pi$
E_1	Electric dipole	1	Yes
M_1	Magnetic dipole	1	No
E_2	Electric quadrupole	2	No
M_2	Magnetic quadrupole	2	Yes
E_3	Electric octupole	3	Yes
M_3	Magnetic octupole	3	No

Selection-Rule Example: ^{23}Na

Consider a ^{23}Na transition from $\frac{7}{2}^+$ to $\frac{3}{2}^+$. **Step 1 – Parity change:** both states have positive parity $\Rightarrow \Delta\pi =$ No. **Step 2 – Allowed l values:**

$$\left|\frac{7}{2} - \frac{3}{2}\right| \leq l \leq \frac{7}{2} + \frac{3}{2} \quad \Rightarrow \quad 2 \leq l \leq 5$$

Step 3 – Classify each allowed l:

l	$\Delta\pi$ required	Type
2	No	E_2
3	No	M_3
4	No	E_4
5	No	M_5

Lowest multipolarities dominate in practice $\Rightarrow$ the transition is predominantly E_2, with progressively smaller M_3, E_4, M_5 contributions.

Summary

Summary (1/2): pp Chains and Burning Stages

- Deuterium burning is dominated by $p + d \to {}^3\mathrm{He} + \gamma$; quasi-equilibrium gives $(D/H)_e \sim 10^{-18}$, in stark contrast to the observed $\sim 10^{-5}$ – a Big-Bang signature
- ${}^3\mathrm{He}$ burning completes pp-I via ${}^3\mathrm{He} + {}^3\mathrm{He} \to {}^4\mathrm{He} + 2p$; protons cannot complete the chain because ${}^4\mathrm{Li}$ is not particle-stable
- Three branches of the pp chain split at ${}^3\mathrm{He}$:
  - pp-I (86%, $Q_{\mathrm{eff}} = 26.2$ MeV)
  - pp-II (14% combined, via ${}^7\mathrm{Be}$ electron capture, $Q_{\mathrm{eff}} = 25.66$ MeV)
  - pp-III (small, via ${}^8\mathrm{B}$, $Q_{\mathrm{eff}} = 19.17$ MeV – largest ν loss)
- Net energy production: $\varepsilon = 13.1\, r_{11}$ MeV cm^{-3} s^{-1}, set entirely by the slow $p + p$ rate
- Only about 2.4% of the Sun's mass is involved in active pp burning

Summary (2/2): Cycles, Shell Model, Gamma Decay

- CN cycle (rate-limited by $^{14}\mathrm{N}(p,\gamma)^{15}\mathrm{O}$) is the dominant H-burning channel above $T_6 \approx 20$, and is critical for C/N isotope synthesis
- CNO bi-cycle adds the $^{15}\mathrm{N}(p,\gamma)^{16}\mathrm{O}$ branch (1000:1 against (p,α)); CNO3 and CNO4 extend to O, F, and Ne isotopes
- Hot CNO (novae, supernovae): β^+-unstable nuclei react with protons before decaying $\Rightarrow$ richly interconnected "many-cycled monster"
- NeNa and MgAl cycles synthesize elements between $^{20}\mathrm{Ne}$ and $^{27}\mathrm{Al}$ – not significant for energy, but produce $^{22}\mathrm{Ne}$ and $^{26}\mathrm{Al}$ found in meteorites; $^{28}\mathrm{Si}$ seeds the next burning stage
- Shell model: harmonic / Woods–Saxon potential with spin–orbit coupling gives (n, l, j) single-particle states; explains magic numbers and supplies J^π, transition strengths
- Gamma decay: $E_\gamma \approx E_i - E_f$; multipolarity determined by $|I_f - I_i| \leq l \leq I_f + I_i$; electric vs. magnetic character set by parity change via $(-1)^l$ vs. $(-1)^{l+1}$